



ANGELA BARONE

MACHINE LEARNING | ENGINEERING DOCTORATE | HUMAN-AI INTERACTION

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SUMMARY

Data scientist/machine learning engineer with **~2 years of working experience** and formal education (doctorate). I design *AI solutions that support better human decisions*.

TECHNICAL INTERESTS: LLMs and LLM post-training, RAG, Knowledge Graph, Agentic AI, Explainable AI, Human-AI Interaction, Big Data, Digital Twins

TECHNICAL SKILLS: PyTorch · TensorFlow · Hugging Face · Git · Docker · Spark / PySpark · MLOps · Azure

WORK EXPERIENCE

Data Scientist, Eindhoven University of Technology

Sep 2025 - Present

As part of my Engineering Doctorate, I have allocated more 50% of my working hours on consulting projects for client companies and provided *research reports* for them:

Project 3 | Client: **Censo** (Green Energy Provider),

DATA ENGINEERING/ML ENGINEERING

ETL Pipeline + hybrid RAG (front end, back end, evaluation). All deployed on Azure.

Collaborated in a team using Git branching workflows.

Project 2 | Client: **Wärtsilä**, Role: Project Team Leader

Transforming unstructured marine service PDFs into a structured dataset for pattern recognition, predictive maintenance and commercial follow up.

LLMs FOR PREDICTIVE MAINTENANCE

- Engineered a LLM inference pipeline using Qwen via Hugging Face Transformers, with pre-summarization for long documents;
- Deployed on the TU/e Umbrella HPC cluster with multi-GPU distribution, reducing processing time from days to hours;
- Designed a quantitative evaluation framework without ground truth labels (unsupervised learning evaluation);
- Supervised the design of a knowledge graph enabling cross-document pattern recognition;
- Collaborated in a team using Git branching workflows

Project 1 | Client: **Forescout Technology** (Cybersecurity)

ANOMALY DETECTION

- Designed semi-supervised (positive-unlabeled) learning pipelines for large-scale asset classification in enterprise networks.
- Built device fingerprints from network telemetry using text embeddings, boosting algorithms (XGBoost), protocol signals, and rule-based features, and used clustering to explore and validate device groupings.
- Increased asset visibility by roughly 60%, using only data already available in the system.*

- Automated end-to-end analysis of employee survey data for the Working Platform team by developing an interactive Power BI dashboard, significantly reducing manual reporting effort.
 - Built a Python-based sentiment analysis pipeline for open-ended survey responses and integrated results into Power BI with clear visualizations for stakeholders.
 - Implemented an automated data pipeline connecting Microsoft Surveys, SQL databases, and Power Automate, enabling real-time data ingestion and reporting.
 - Translated qualitative feedback into actionable insights to support data-driven decision-making within a banking context.
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EDUCATION

Engineering Doctorate - TU/e (Eindhoven University of Technology)

Sep 2025- Present

Engineering Doctorate in Data Science

The engineering doctorate works as a bridge between industries and academia, allowing in-company research and traditional post-master education. Besides the training projects, I am currently working on writing academic papers. Currently working on:

- LLMs biases and Human AI interaction benchmarked against professional debates

Master of Science in Data Science, Tilburg University

Jan 2024 - June 2025

Average grade: 8.78 (*Cum Laude*)

- Thesis: 8.7 (Top 5 Thesis of the DSS Master's Program, selected for the Best Thesis Symposium)
 - Selected courses (*more than the regular ECTs*): Machine Learning (9.0), Deep Learning (8.5), Big Data, Statistics (9.5), Data Mining for Businesses and Governance (7.5), Spatiotemporal Data Analysis (9.0), Image Analysis (8.5), Regulation & Law (10), Sustainability, Privacy and Security (9.5), R, Advanced Python, SQL.
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ADDITIONAL INFORMATION

- **Languages:** English, Italian, Spanish (B2), French (B2), Dutch (A2+, Certified)
- **Certifications:** Databricks (ongoing), Azure (ongoing)
- **Awards/Activities:** Selected for best thesis symposium (best 5 thesis out of 500), Winner (2nd place) of a Hackathon organized by Nestle' and Twente University, finalist for bunq's Hackathon on **Agentic AI**
- **Hobbies:** Gardening, Interior Design, Photography, History